

ZANESVILLE, OH, USA

WMO: 724286

Lat: 39.944N

Lon: 81.892W

Elev: 880

StdP: 14.23

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93824

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.3	(c) 9.2	(d) -4.9	(e) 4.4	(f) 5.5	(g) 0.3	(h) 5.8	(i) 10.9	(j) 23.4	(k) 31.8	(l) 20.3	(m) 32.8	(n) 5.1	(o) 260	(p) 0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.5	(c) 89.8	(d) 73.7	(e) 87.5	(f) 72.8	(g) 85.2	(h) 71.5	(i) 76.5	(j) 85.9	(k) 75.0	(l) 83.5	(m) 73.6	(n) 81.5	(o) 8.0	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	128.5	81.3	72.2	123.3	79.7	71.0	118.2	78.2	40.4	86.4	38.9	83.7	37.6	81.8	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.0	(b) 16.9	(c) 15.2	(e) -4.9	(f) 92.9	(g) 8.0	(h) 3.1	(i) -10.7	(j) 95.1	(k) -15.3	(l) 96.9	(m) -19.8	(n) 98.7	(o) -25.6	(p) 100.9		
(4) 19.0	(5) 16.9	(6) 15.2	(7) -4.9	(8) 92.9	(9) 8.0	(10) 3.1	(11) -10.7	(12) 95.1	(13) -15.3	(14) 96.9	(15) -19.8	(16) 98.7	(17) -25.6	(18) 100.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.6	29.2	32.5	41.0	52.7	62.1	70.2	73.5	72.3	65.7	54.1	43.0	34.3
	DBStd	17.92	12.60	11.47	10.89	9.23	7.94	5.98	4.91	4.87	6.96	8.50	9.38	10.15
	HDD50	2361	654	501	323	77	6	0	0	1	53	250	496	
	HDD65	5366	1109	912	745	380	154	23	2	74	352	659	951	
	CDD50	3327	10	10	44	157	380	605	728	690	471	180	42	10
	CDD65	856	0	0	1	10	63	178	265	230	95	14	0	0
	CDH74	7403	0	1	19	134	618	1512	2263	1879	845	129	3	0
	CDH80	2237	0	0	2	16	143	462	760	587	249	18	0	0
	WSAvg	6.2	7.9	7.7	7.6	7.5	6.0	5.4	4.7	4.3	4.5	5.6	6.6	7.1
	PrecAvg	38.8	2.4	2.2	3.4	3.6	4.0	3.9	4.3	3.7	3.0	2.4	3.1	2.8
	PrecMax	54.4	5.9	4.2	9.9	6.9	8.0	13.0	8.1	10.6	6.2	5.6	13.3	7.3
	PrecMin	26.5	0.5	0.3	1.0	0.8	1.5	0.7	1.2	0.6	0.4	0.2	0.7	1.0
	PrecStd	6.1	1.2	1.1	1.8	1.6	1.7	2.5	2.1	2.2	1.6	1.3	2.0	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.3	67.8	77.1	82.4	87.0	90.9	93.1	91.6	90.4	83.0	72.6	65.2
		MCWB	56.8	56.1	61.1	63.4	70.8	73.1	76.3	75.3	71.4	68.3	60.1	58.1
	2%	DB	58.9	60.9	70.6	78.0	83.8	87.9	89.7	88.4	86.5	77.9	67.2	59.3
		MCWB	54.5	52.4	57.1	61.2	68.1	72.6	74.8	73.6	70.8	64.8	57.4	53.7
	5%	DB	53.8	55.3	64.7	73.9	81.1	85.2	87.0	85.9	82.3	73.6	63.1	55.0
		MCWB	50.0	49.1	53.8	58.9	67.1	71.3	73.5	72.3	68.6	62.6	54.6	50.4
	10%	DB	47.5	49.9	59.4	69.7	77.4	82.4	84.3	83.3	78.9	69.5	58.9	50.2
		MCWB	42.9	44.0	51.6	56.5	65.3	70.0	72.1	71.0	67.1	60.9	52.2	46.2
	0.4%	WB	58.7	59.0	62.3	67.0	73.3	76.7	78.4	78.3	74.9	71.1	63.3	59.5
		MCDB	62.5	66.1	73.9	78.6	83.4	86.7	89.0	87.6	84.9	81.4	69.3	63.9
	2%	WB	55.1	54.6	59.1	63.5	70.7	74.5	76.8	76.3	72.9	67.8	59.3	55.1
		MCDB	58.6	59.1	67.4	73.5	80.1	83.8	86.6	84.9	81.6	74.2	65.3	58.3
(31) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	50.5	49.9	56.0	61.1	68.8	73.1	75.3	74.6	71.1	64.7	56.2	51.1
		MCDB	53.6	54.9	63.0	70.2	77.7	82.4	83.7	82.1	78.8	71.2	61.6	54.2
	10%	WB	43.6	44.2	52.2	58.9	66.8	71.7	73.8	72.9	69.3	61.8	52.9	46.0
		MCDB	46.9	49.4	58.7	66.8	75.0	80.2	81.5	79.7	76.0	68.1	58.4	50.1
(35) Mean Daily Temperature Range	5% DB	MDBR	15.4	16.7	19.3	22.2	21.2	20.0	19.5	20.2	21.5	20.9	18.3	14.5
		MCDBR	18.6	24.5	27.4	27.8	25.5	22.9	21.8	22.9	25.1	25.9	24.3	20.3
		MCWBR	15.2	18.3	16.7	15.2	12.6	10.5	9.3	9.9	11.4	14.5	16.5	16.8
		MCDBR	17.6	22.4	22.9	22.7	21.3	19.7	19.1	19.1	20.1	21.5	19.9	18.8
	5% WB	MCWBR	15.3	18.5	15.5	14.1	11.4	9.8	9.0	9.3	9.9	13.4	15.6	17.1
		taub	0.308	0.326	0.351	0.390	0.432	0.459	0.469	0.461	0.399	0.364	0.323	0.304
		taud	2.452	2.387	2.380	2.290	2.191	2.159	2.148	2.153	2.304	2.402	2.494	2.499
		Ebn at Noon	272	282	286	280	270	262	258	256	266	265	265	266
(44) All-Sky Solar Radiation	RadAvg	511	764	1094	1474	1721	1882	1832	1702	1392	953	647	436	
	RadStd	49	106	100	139	143	137	114	95	129	92	89	52	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	N/A
(47) Regional (42 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+232	+124

Nomenclature: See separate page